

MO YUNXIANG

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EDUCATION

The Hong Kong University of Science and Technology

Sep 2024 – Present

BSc, Double Major in Computer Science & Mathematics; Extended Major in Artificial Intelligence; Year 2

GPA: 4.03 / 4.3 Relevant Coursework: Algorithms, Statistical Inference, Machine Learning, Linear Algebra, Probability Theory

A+ in all Mathematics and Computer Science courses (top 5% in each course)

Honors:

- **S.S. Chern Class** [link] – HKUST Department of Mathematics, 2025. *Honor for top academic performance across all mathematics coursework.*
- **University's Scholarship Scheme for Continuing Undergraduate Students** (Top 1%), 2024, 2025
- **Dean's List Honor** (twice; GPA above 3.7), 2024, 2025

RESEARCH EXPERIENCE

First-author paper, ACL 2026 Main Conference (accepted; oral pending) [OpenReview]

“DixitWorld: Evaluating Multimodal Abductive Reasoning in Vision-Language Models with Multi-Agent Dixit Gameplay”

Yunxiang Mo, Tianshi Zheng, Qing Zong, Jiayu Liu, Baixuan Xu, Yauwai Yim, Chunkit Chan, Jiaxin Bai, Yangqiu Song. Advisor: Prof. Yangqiu Song.

- Extended version of our EMNLP 2025 Workshop paper (below); adds a Medium difficulty tier (252 vs. 168 QA items), 72B-parameter scaling ablation, and calibration/sensitivity analyses
- Sharper headline: 78% of storyteller rounds yield zero points while best listener accuracy reaches 75.6%; $r = 0.947$ correlation between Arena and Bench validates structural nature of the gap
- AC meta-review 9/10 (Strong Accept); recommended for main-conference oral presentation

First-author paper, EMNLP 2025 Workshop (Spotlight) [arXiv]

“DixitWorld: Evaluating Multimodal Abductive Reasoning in Vision-Language Models with Multi-Agent Dixit Gameplay”

Yunxiang Mo, Tianshi Zheng, et al., Yangqiu Song.

- Proposed DixitArena (Dixit-inspired multi-agent environment) and DixitBench (static QA benchmark) to evaluate multimodal abductive reasoning in vision-language models
- Identified a structural storyteller–listener asymmetry: small open-source models excel at creative clue generation, while large proprietary models excel at hypothesis selection
- Implementation in PyTorch with Transformer-based multimodal architectures

INTERNSHIP EXPERIENCE

Beijing Ingenic Semiconductor Co., Ltd.

Jun 2025 – Aug 2025

Machine Learning Engineer Intern

- Developed and optimized machine learning models for embedded and on-chip AI scenarios
- Participated in model training, evaluation, and inference pipeline development using PyTorch
- Assisted in deploying ML models to edge devices with constraints on latency and memory
- Collaborated with algorithm and hardware teams to align model design with chip-level constraints

Benchmark Architectural Design Co., Ltd

Jan 2025

- Developed front-end modules using the MFC framework for a mini-program project
- Implemented UI design, event handling, and system debugging in a team environment

TECHNICAL SKILLS

Machine Learning & NLP: Transformer Architectures, Multimodal Reasoning, Abductive Inference, Prompt Engineering, Fine-tuning

Frameworks & Tools: PyTorch, Hugging Face Transformers, OpenAI API

Programming Languages: Python, C++, MATLAB

Languages: Chinese (Native), English (Fluent)

COMPETITIONS

- National Mathematical Olympiad (China) – Second Prize, 2023
- National Informatics Competition (China) – Second Prize, 2022